

From Metaphor to Identification

A Causal Architecture for the Extended Phenotype Theory of Law

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August 2026

Abstract

Evolutionary accounts of law frequently use causal language while leaving the relevant causal questions underspecified. Legal norms are said to produce institutions, institutions to select norms, and reforms to fail because of lock-in, yet persistence, prediction, intervention, mediation, and counterfactual attribution are often treated as interchangeable forms of explanation. This article develops the Dynamic Extended-Phenotype Causal Architecture (DEPCA), a structural framework for converting the Extended Phenotype Theory of Law from a mechanism-rich research program into an identification-ready one. DEPCA represents legal replicators, institutional configurations, agent behavior, social outcomes, selective environments, and subsequent replicator retention as time-indexed variables in a dynamic structural causal model. Temporal unrolling preserves reciprocal causation without violating the acyclicity required by ordinary causal graphs. Six propositions follow. First, institutional feedback can be represented as cross-period causation rather than a same-time causal circle. Second, institutions may be outcomes, mediators, moderators, or selection mechanisms, and those roles imply different estimands. Third, indices of rigidity or fitness are not identical to the latent causal properties they seek to measure. Fourth, the ladder of causal reasoning and levels of intentionality are orthogonal: an agent's cognitive sophistication does not determine the causal question an analyst may identify. Fifth, evolutionary persistence and causal efficacy are distinct. Sixth, causal attribution cannot by itself establish legal responsibility, validity, or remedy. Applications to constitutional lock-in, AI liability, and channel opacity show how DEPCA changes research design and limits permissible claims. The article concludes with a preregistration-ready protocol for longitudinal, quasi-experimental, mediation, and transportability studies. The contribution is theoretical and methodological. It does not claim that any particular legal intervention has already been causally identified.

Keywords: causal inference; structural causal models; extended phenotype; legal evolution; institutional lock-in; counterfactuals; mediation; legal responsibility

1. Introduction

Evolutionary explanations of law are strongest when they identify a mechanism and weakest when they allow the vocabulary of mechanism to substitute for evidence of causation. A legal doctrine persists, so it is described as fit. A professional institution reproduces the doctrine, so it becomes the doctrine's extended phenotype. Reform fails, so institutional lock-in is invoked as its cause. Courts, regulated actors, and administrative agencies then alter the environment in which the doctrine competes, and the process is characterized as reciprocal causation. Each statement may be plausible. None, by itself, specifies what would have happened under a different institutional arrangement, which variables must be controlled, which must not be controlled, or whether the proposed effect can be learned from the available observations.

This gap is not unique to evolutionary legal theory. Modern causal inference developed in response to the same problem across epidemiology, economics, statistics, and artificial intelligence: statistical regularity does not answer an intervention question unless substantive assumptions connect the observed distribution to a causal model. Structural causal models distinguish seeing from doing and

doing from imagining otherwise (Pearl 2009; Pearl and Mackenzie 2018). The potential-outcomes tradition frames the same discipline around well-defined interventions, counterfactual outcomes, exchangeability, positivity, consistency, and time (Hernán and Robins 2020). These traditions differ in notation and emphasis, but converge on a central demand. Before estimating an effect, one must state the causal question.

The Extended Phenotype Theory of Law (EPT-Law) needs that demand precisely because its ontology is causal. In biological extended-phenotype analysis, the question is not whether a beaver dam correlates with beaver genes. It is whether variation in replicators affects structures beyond the organism through a traceable chain, and how those structures alter subsequent survival and reproduction (Dawkins 1982). Applied to law, the theory proposes that legal replicators express themselves through courts, professional organizations, enforcement routines, curricula, classifications, and infrastructures that extend their persistence and behavioral reach (Lerer 2026a). Niche-construction analysis adds feedback: institutions do not merely express inherited norms but modify the selective environment facing later normative variants (Odling-Smee, Laland, and Feldman 2003; Laland et al. 2015; Lerer 2026b). A research program built around extended effects and reciprocal causation cannot remain agnostic about causal identification indefinitely.

The present article supplies an architecture rather than an estimated effect. Its central contribution is the Dynamic Extended-Phenotype Causal Architecture (DEPCA), a time-indexed structural model that separates six elements: legal replicators, institutional configurations, agent behavior, social outcomes, selective environments, and subsequent replicator retention. The architecture does four kinds of work.

First, it converts reciprocal causation from a verbal circle into a sequence of causal questions. A norm at time t may shape an institution at t ; that institution may shape behavior and outcomes; those outcomes may alter the selective environment; and the changed environment may affect which norm is retained or adopted at $t + 1$. Feedback is real, but it is feedback through time.

Second, DEPCA distinguishes the role assigned to an institution. The same court system cannot be treated indiscriminately as an outcome of a doctrine, a mediator of its behavioral effect, a moderator of reform efficacy, and a confounder to be adjusted away. Each role produces a different estimand and a different adjustment strategy.

Third, the architecture introduces a measurement firewall. Constitutional rigidity, citation centrality, institutional hysteresis, and memetic fitness are latent or constructed properties. An index may be a useful indicator, but a score is not automatically the causal property itself. Predicting reform outcomes with an index does not establish that changing the index would change the outcome.

Fourth, it preserves a legal-normative firewall. A causal model may estimate whether a design contributed to harm, whether an audit would have prevented it, or whether a reform altered litigation. It cannot, without additional legal premises, decide whether a person owed a duty, whether conduct was culpable, whether a norm is valid, or which remedy follows. Law depends on causation, but legal judgment is not reducible to it.

The argument proceeds as follows. Section 2 locates the identification gap within evolutionary and institutional legal research. Section 3 defines causal questions and explains why temporal order is indispensable. Section 4 presents DEPCA and six propositions. Section 5 develops identification strategies. Section 6 applies the architecture to constitutional lock-in, AI liability, and channel opacity. Section 7 separates causal from normative inference. Section 8 gives an empirical research

protocol. Section 9 considers rival explanations and limitations. Section 10 states a research agenda and distinctive predictions. Section 11 concludes.

2. The Identification Gap in Evolutionary Legal Theory

2.1 From analogy to mechanism

The appeal of evolutionary analysis in law is easy to understand. Legal orders display variation, differential persistence, inheritance through education and precedent, competition among doctrines, path dependence, and adaptation to changing environments. Legal rules can also create structures that outlive their designers. Specialized courts, professional credentials, administrative routines, standard forms, databases, and enforcement budgets may continue reproducing a normative pattern after the political coalition that introduced it has disappeared. The extended-phenotype vocabulary provides a compact account of this causal reach.

The danger is equally clear. Evolutionary terminology can redescribe an outcome without explaining it. If every persistent rule is called fit, persistence becomes both explanandum and evidence. If every institution associated with a rule is called its phenotype, the direction of influence is assumed rather than tested. If failure of change is labeled lock-in only after failure occurs, the construct lacks prospective content. Theoretical fertility then produces a research program rich in plausible narratives but weak in discriminating tests.

EPT-Law has already moved beyond pure analogy in three respects. It specifies agent-level strategic mechanisms through evolutionary game theory; proposes measurable institutional constructs; and treats institutions as both phenotypic expressions and constructed niches (Lerer 2026a, 2026b). These advances make the causal gap more visible, not less. A mechanism is a candidate causal structure. A measure is a candidate operationalization. A prediction is a candidate test. None becomes an identified effect merely by being formalized.

Consider a high score on a Constitutional Lock-in Index and a subsequent reform failure. At least five interpretations are possible. The score may measure a genuine set of institutional causes. It may summarize consequences of an earlier cause, such as durable political conflict. It may proxy for omitted factors, such as state capacity or coalition fragmentation. It may partly incorporate information produced after treatment, thereby leaking the outcome into the predictor. Or it may predict well in one set of jurisdictions while failing under transport because the components have different meanings elsewhere. A causal architecture must keep these possibilities open until design and evidence distinguish them.

2.2 Persistence, fitness, and efficacy

Evolutionary fitness concerns differential replication or retention relative to a specified environment. Causal efficacy concerns the difference an intervention would make to an outcome. They are related but not identical. A doctrine may persist because it is bundled with a highly transmissible professional identity, even if its direct effect on compliance is small. Conversely, a short-lived emergency norm may have a large causal effect during its brief operation. Longevity is not a universal measure of causal force.

The distinction matters because legal scholarship often moves in both directions without notice. It infers efficacy from persistence: the rule survived, therefore it shaped conduct. Or it infers fitness from efficacy: the rule changed behavior, therefore it will be retained. Both inferences require additional mechanisms. Persistence may result from documentary inertia, agenda control, or absence of a revising institution. Behavioral efficacy may provoke backlash and accelerate repeal.

The relevant selection environment must be stated, as must the outcome whose causal response is at issue.

This is also where cultural-attraction theory poses a productive challenge. Cultural transmission need not consist of high-fidelity copying by stable replicators. Representations may be repeatedly reconstructed toward psychological or institutional attractors (Sperber 1996; Claidière, Scott-Phillips, and Sperber 2014). Law is unusually favorable to high-fidelity transmission because canonical texts, citation rules, professional training, and coercive institutions stabilize tokens. Yet interpretation and application remain reconstructive. DEPCA does not resolve the replicator-attractor dispute by definition. It allows rival models to encode different mechanisms and derive competing observable implications.

2.3 Reciprocal causation is not exemption from identification

Niche-construction theory correctly resists a one-way picture in which organisms or norms merely adapt to an independent environment. Organisms modify environments; institutions modify the incentives, categories, information channels, and enforcement structures facing later actors. Legal evolution is therefore path-dependent and endogenous.

But endogeneity is not a license to draw arrows in every direction. A graph in which norms cause institutions and institutions cause norms at the same analytical instant does not identify a causal process. It states that the variables belong to a system. That may be ontologically correct and methodologically insufficient.

The solution is temporal and modular. A doctrine inherited at the beginning of a period shapes institutional decisions during that period. Those decisions alter behavior and outcomes. The resulting political, epistemic, fiscal, and professional environment shapes which variants are transmitted into the next period. Longer feedback can be represented across additional lags. The model remains open to reciprocal influence while requiring the researcher to state when each variable is determined.

2.4 The target trial hidden inside legal reform

Every credible causal claim about reform contains an implicit experiment. The analyst imagines comparable jurisdictions, institutions, cases, or periods, some exposed to a legal change and others not exposed, and asks how outcomes would differ. Hernán and Robins (2020) call the explicit version a target trial: eligibility, treatment strategies, assignment, follow-up, outcome, causal contrast, and analysis are specified before observational data are used to emulate it.

Legal reforms rarely permit literal randomization, but the target-trial discipline remains valuable. “Does constitutional rigidity cause reform failure?” is too vague. A better question is: among labor-market reform proposals eligible for enactment in a defined set of presidential systems between specified dates, what would be the five-year probability of sustained implementation if a severable constitutional-review procedure were available compared with the probability under the existing review architecture? The reform, population, horizon, outcome, and intervention are now contestable but visible.

3. Causal Questions, Not Causal Vocabulary

3.1 Three rungs and several estimands

Pearl and Mackenzie (2018) popularize a ladder with three levels: association, intervention, and counterfactuals. At the first level, the analyst asks what is observed among units with a feature. At

the second, the analyst asks what would happen under an action that sets a variable. At the third, the analyst asks what would have happened to the same unit or event under an alternative history.

Legal research needs all three. Association can reveal which doctrines co-occur with high reversal rates. Intervention analysis can estimate the average effect of changing a review rule. Counterfactual analysis can address whether a particular plaintiff would probably have avoided harm under a different warning or audit. The evidence and assumptions become stronger as the question rises. A dataset adequate for prediction may be inadequate for intervention; a population effect may be inadequate for singular attribution.

Within those rungs, further distinctions matter. The average treatment effect compares average potential outcomes under two interventions. The treatment effect among treated units answers a different question (Imbens and Rubin 2015). A controlled direct effect fixes a mediator; a total effect allows mediation to operate. Probabilities of necessity and sufficiency address forms of attribution relevant to legal disputes but generally require stronger assumptions or bounds (Pearl 2009). Transportability asks whether an effect learned in one population can be carried to another whose causal mechanisms or distributions differ (Pearl and Bareinboim 2014).

3.2 Variables and structural equations

Let a structural causal model be a tuple containing endogenous variables, exogenous variables, structural functions, and a probability distribution over exogenous variables. Each endogenous variable is determined by its parents and an exogenous disturbance. The graph encodes exclusions and causal order, not merely statistical dependence.

For EPT-Law, define:

- R_t : the distribution or content of legal replicators available at time t ;
- I_t : institutional configuration, including courts, agencies, professional bodies, routines, and technical infrastructures;
 - A_t : behavior of relevant agents;
- O_t : legal, social, distributive, or administrative outcomes;
- S_t : the selective environment affecting transmission and retention;
 - H_t : inherited history and prior institutional state;
 - X_t : observed contextual covariates;
- U_t^k : unobserved causes for each structural equation.

A minimal dynamic specification is:

$$\begin{aligned}
 I_t &= f_I(R_t, H_t, X_t, U_t^I), \\
 A_t &= f_A(I_t, R_t, X_t, U_t^A), \\
 O_t &= f_O(A_t, I_t, R_t, X_t, U_t^O), \\
 S_{t+1} &= f_S(O_t, I_t, A_t, X_{t+1}, U_{t+1}^S), \\
 R_{t+1} &= f_R(R_t, S_{t+1}, I_t, H_t, U_{t+1}^R).
 \end{aligned}$$

The equations are deliberately nonparametric. They state a causal architecture without pretending that the functions or distributions have been estimated. A specific study must replace broad variables with measured constructs, choose lags, and justify exclusions.

3.3 Seeing, doing, and changing legal institutions

Conditioning on $I_t = i$ asks about units observed with an institutional configuration. Intervening with $do(I_t = i)$ replaces the structural equation for I_t and asks what outcomes would follow if the configuration were set. Those quantities differ whenever common causes influence both institutions and outcomes.

This difference is central to comparative law. Countries with specialized constitutional courts differ from countries without them in history, party systems, legal culture, state capacity, and patterns of conflict. Observing different reform outcomes does not identify the effect of creating such a court. Even statistical adjustment is valid only if the chosen variables block the relevant back-door paths without conditioning on mediators or colliders.

Legal interventions also have versions. “Adopt judicial review” may mean decentralized review, a centralized constitutional court, abstract review before enactment, concrete review after injury, weak-form declarations, or strong invalidation. Consistency fails if heterogeneous versions are collapsed into one treatment while producing materially different outcomes. DEPCA therefore requires the intervention to be institutionally executable, not merely conceptually attractive.

4. The Dynamic Extended-Phenotype Causal Architecture

4.1 Core architecture

DEPCA treats a legal extended phenotype as a mediated, temporally extended causal process. A replicator is not identified by persistence alone. The researcher specifies a norm, doctrine, classification, interpretive rule, or memplex whose variation can be traced into institutions or behavior. The institutional phenotype is the set of structures through which that replicator affects transmission, compliance, enforcement, or later selection. The selective environment is not a residual category; it is a set of variables through which current outcomes influence future replication.

The core process can be read from left to right within a period and then across periods:

$$R_t \rightarrow I_t \rightarrow A_t \rightarrow O_t \rightarrow S_{t+1} \rightarrow R_{t+1}.$$

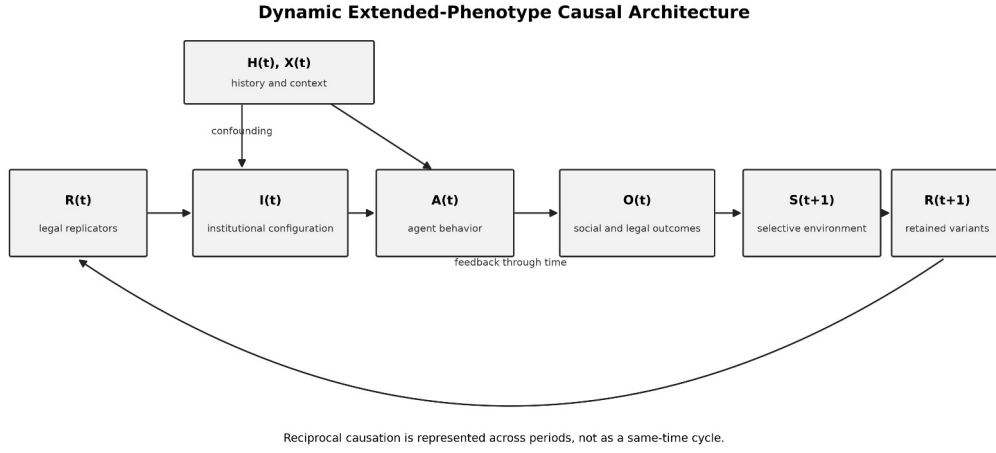


Figure 1. The Dynamic Extended-Phenotype Causal Architecture. Legal replicators affect institutional configuration, behavior, and outcomes; those consequences reshape the selective environment and later retention. History and context may confound several relations.

Additional arrows are expected. Replicators may directly influence agents through focal meaning; institutions may directly affect outcomes; history may influence every stage; and external shocks may alter both behavior and selection. The value of the architecture is not graphical minimalism. It is the requirement that every added arrow correspond to a proposed mechanism and every missing arrow to an exclusion assumption.

4.2 Proposition 1: Temporal Unrolling

Proposition 1 (Temporal Unrolling). If reciprocal influence between legal replicators and institutions operates with a nonzero temporal lag, the system can be represented as an acyclic graph over time-indexed variables without denying reciprocal causation.

Argument. Suppose R_t affects I_t , while institutional consequences affect which replicators are retained. If retention occurs after institutional operation, the reverse influence is $I_t \rightarrow R_{t+1}$, not $I_t \rightarrow R_t$. Ordering variables by time produces no directed path from a later variable to an earlier one. The unrolled graph is acyclic even though the rolled system contains feedback. This is the standard logic of dynamic causal systems: recurrence across periods does not require simultaneous causal cycles.

The proposition is modest but consequential. It turns “law and institutions co-constitute each other” into a family of lagged claims. It also reveals where simultaneity may remain. If judicial interpretation and doctrinal content are defined within the same decisional act, temporal order may be too fine to observe. The researcher must then redefine variables, model simultaneous equations under additional assumptions, or withhold identification.

4.3 Proposition 2: Institutional Role Differentiation

Proposition 2 (Institutional Role Differentiation). The causal effect attributed to a legal replicator is not well-defined until the role of institutions as outcome, mediator, moderator, selection mechanism, or confounder is specified.

Argument. Consider a constitutional clause R , specialized labor courts I , employer compliance A , and worker outcomes O . If the question is whether the clause created specialized courts, I is the outcome. If the question is the total effect of the clause on outcomes, I may mediate that effect and should ordinarily not be adjusted away. If the question is whether the clause works differently in

countries with specialized courts, I is a moderator. If preexisting court capacity affected both clause adoption and later outcomes, a prior measure of I may confound the comparison. These roles cannot be merged without changing the estimand.

The proposition corrects a recurrent legal-research error: “controlling for institutions” may remove the very pathway through which law operates. Conversely, failing to control for preexisting institutional causes may attribute historical capacity to the new norm. Temporal indexing distinguishes I_{t-1} , a possible confounder, from I_t , a possible mediator.

4.4 Proposition 3: Measurement Non-Identity

Proposition 3 (Measurement Non-Identity). An index that predicts an institutional outcome is not thereby identical to, or an identified intervention on, the causal construct it is designed to measure.

Argument. Let L denote latent institutional lock-in and M an observed index built from precedent density, doctrinal elaboration, professional organization, and enforcement infrastructure. Even if M predicts reform failure, at least three structures remain possible: $L \rightarrow M$ and $L \rightarrow Y$; component causes C_j independently affect both M and Y ; or prior outcomes affect components included in M , producing reverse causation. An intervention $do(M = m)$ is usually meaningless because changing a reported score while leaving components unchanged does nothing. The meaningful interventions target components or mechanisms.

The implication is constructive. Indices such as CLI or JurisRank may be excellent risk stratifiers, measurement instruments, or summaries of state. To support policy, the analyst must decompose them into manipulable variables and estimate component-specific or bundled intervention effects. Reliability, predictive validity, construct validity, and causal validity are distinct achievements.

4.5 Proposition 4: Orthogonality of Causal and Intentional Ladders

Proposition 4 (Orthogonality). Levels of causal inquiry and levels of agent intentionality are analytically independent dimensions.

Argument. The causal ladder classifies questions an analyst asks: observation, intervention, or counterfactual. An intentionality hierarchy classifies the recursive mental-state architecture attributed to an agent. A zero- or first-order optimizer can be the subject of a sophisticated counterfactual inquiry. A highly reflective human can be studied only associationally when the data do not identify an intervention. Therefore neither ladder fixes the level of the other.

This matters for AI governance. An AI system need not possess human-like counterfactual consciousness for researchers to ask the counterfactual effect of deploying it. Conversely, a model’s capacity to generate counterfactual language does not establish that a study of its legal effects has identified a counterfactual quantity. Conflating the two dimensions would turn a methodological taxonomy into a theory of mind.

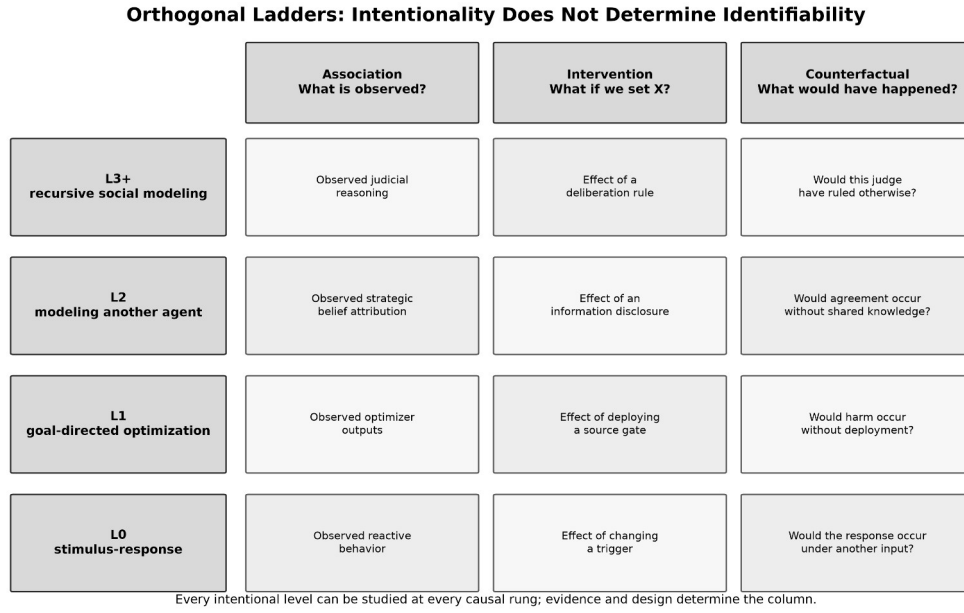


Figure 2. The causal ladder and intentionality levels are orthogonal. Every class of agent can be studied through associational, intervention, or counterfactual questions when the evidence and assumptions permit.

4.6 Proposition 5: Persistence-Efficacy Separation

Proposition 5 (Persistence-Efficacy Separation). Differential persistence of a legal replicator neither entails nor is entailed by a large causal effect on a specified social outcome.

Argument. Let fitness F_R be the expected rate of transmission or retention of replicator R in a defined environment, and let τ_R be the causal effect of adopting R on outcome O . No general monotonic relation follows. A rule may have F_R high because it is easily cited, professionally rewarded, or insulated from revision while τ_R is near zero. Another rule may have a large short-term τ_R yet low F_R because the affected coalition repeals it. Any relation between fitness and efficacy is an empirical hypothesis mediated by selection mechanisms.

This proposition protects EPT-Law from circularity. Fitness must be defined relative to replication, and outcome effects must be estimated separately. The framework may then study whether particular kinds of efficacy improve or reduce fitness under specified environments.

4.7 Proposition 6: Causal-Normative Separation

Proposition 6 (Causal-Normative Separation). No descriptive causal conclusion alone entails legal responsibility, normative validity, or a remedy.

Argument. A structural model may establish that intervention X increased the probability of harm Y , or that without X a particular harm probably would not have occurred. Liability nevertheless depends on additional premises: duty, protected interest, scope of responsibility, standard of conduct, attribution to a legal person, defenses, burden and standard of proof, and remedial rules. Likewise, the persistence or social efficacy of a norm does not establish its validity. The inference from cause to legal consequence requires a normative bridge supplied by positive law and legal argument.

The proposition does not relegate causal analysis to irrelevance. It locates its authority. Better causal analysis can discipline factual attribution and policy prediction. It cannot replace adjudication.

4.8 Selection as a causal process

Selection language introduces a further ambiguity. In one sense, selection is an observed change in the frequency of variants. In another, it is the causal process through which variant properties affect replication. The first can be described from longitudinal distributions. The second requires a model connecting properties, environments, transmission, and retention.

Let $q_t(r)$ be the frequency of replicator variant r at time t , and let $W_t(r, s)$ be its expected transmission or retention under selective environment s . A stylized update is:

$$q_{t+1}(r) = \frac{q_t(r) W_t(r, S_{t+1})}{\sum_j q_t(j) W_t(j, S_{t+1})} + \mu_t(r),$$

where $\mu_t(r)$ represents innovation, reconstruction, migration, or coding error not captured by differential retention. This equation does not establish that law follows biological replicator dynamics. It forces four questions that evolutionary prose often hides. What counts as a variant?

What event counts as reproduction or retention? Which environment determines differential transmission? How much change results from selection rather than introduction or transformation?

In law, the unit may be a textual clause, an interpretive proposition, a citation pattern, a compliance routine, or a broader memplex. These units have different reproduction events. A clause is copied into legislation; a proposition is adopted in a judgment; a routine is repeated in administration; a memplex is reconstructed across institutions. Pooling them creates denominator problems. A paper should select one operational unit for each analysis and state how variants are individuated.

Selection also creates conditioning risks. If a study includes only doctrines that survived long enough to reach a high court, survival is a collider potentially affected by both doctrinal properties and political support. Associations among selected doctrines need not hold in the full population of proposed variants. The same applies to “landmark cases,” published sanctions, and successfully enacted reforms. An evolutionary study must model entry into the observable corpus.

Finally, selection at one level may alter causal effects at another. A compliance rule can reduce present harm while generating professional resistance that lowers future retention. A norm can benefit a coalition while reducing system-level welfare. Multilevel vocabulary becomes causally useful when levels are linked by explicit outcomes, transmission events, and cross-level pathways, not when “the system” is assigned fitness as a single residual property.

4.9 DEPCA as a family of models

DEPCA is not one universal DAG. It is a grammar for constructing domain-specific models. Three common templates illustrate the difference.

The **expression template** asks how variation in a norm produces an institutional phenotype: $R_t \rightarrow I_t$. Its outcomes include creation of a court, budget allocation, professional specialization, or software infrastructure. This template is closest to the biological extended-phenotype claim.

The **operation template** asks how an inherited norm and its institutions affect conduct and consequences: $R_t, I_t \rightarrow A_t \rightarrow O_t$. It supports policy and behavioral questions. Here institutions often mediate or modify the norm’s effect.

The **selection template** asks how outcomes and institutions reshape future transmission: $O_t, I_t \rightarrow S_{t+1} \rightarrow R_{t+1}$. It supports claims about niche construction, persistence, and evolutionary feedback.

A complete study may combine templates, but it should identify a primary one. Otherwise the paper can drift among institutional origin, present effect, and future retention while appearing to answer one question.

5. Identification Strategies for Legal Evolution

5.1 Back-door adjustment and historical confounding

The simplest causal diagrams identify a set of pre-treatment common causes that, if measured and adjusted, block noncausal paths between intervention and outcome. Legal history makes this difficult. Institutional reforms are selected responses to crisis, coalition structure, prior litigation, fiscal capacity, ideology, international pressure, and expected judicial reaction. Many of those variables also affect outcomes.

A back-door strategy is credible only when the substantive graph supports it. Adding all available covariates can increase bias by controlling for colliders or post-treatment variables. For example, litigation volume after reform may mediate the reform's effect or may be jointly caused by enforcement and harm. Adjusting for it without a graph may destroy part of the effect or open a spurious path.

5.2 Quasi-experiments

Legal changes often create discontinuities across time, territory, eligibility, institutional jurisdiction, or case assignment. Difference-in-differences can estimate effects when treated and comparison units would have followed parallel trends absent treatment, no anticipatory behavior invalidates timing, and treatment spillovers are addressed. With staggered timing and heterogeneous effects, modern group-time estimands are preferable to an unexamined two-way fixed-effects coefficient (Callaway and Sant'Anna 2021). Event studies can inspect dynamics but do not rescue a design with differential trends. Regression discontinuity may be appropriate when a legal threshold deterministically or probabilistically assigns treatment and units cannot precisely manipulate the score. Synthetic control can construct a comparison for a singular jurisdiction when pre-treatment fit and donor-pool assumptions are credible.

Each method should be derived from the causal question, not chosen because a dataset accommodates it. A constitutional amendment applying nationwide at one instant may provide interrupted time-series variation but little protection against concurrent shocks. Staggered reforms may offer comparative leverage but introduce heterogeneous-treatment and timing problems. Court-panel assignment may act as an instrument for doctrine exposure only if assignment is as-if random and affects outcomes solely through the relevant doctrinal treatment, an exclusion that is often contestable.

5.3 Longitudinal treatments and feedback

DEPCA's central variables evolve. Institutions at one time affect later exposure, while earlier outcomes affect later institutional choices. Standard regression adjustment may fail when time-varying confounders are themselves affected by prior treatment. Longitudinal g-methods, including the g-formula, inverse-probability weighting, and structural nested models, were developed for this structure (Robins 1986; Hernán and Robins 2020).

An EPT-Law study of repeated reform attempts illustrates the problem. Prior reform failure may increase professional mobilization, change precedent density, and influence the design of the next proposal. Those variables confound the effect of the next proposal while being consequences of earlier proposals. Treating each attempt as independent or adjusting with ordinary regression can produce biased estimates. A longitudinal design must define dynamic treatment regimes and histories.

5.4 Mediation

Extended-phenotype claims are frequently mediation claims in disguise. A doctrine affects institutional structure, which affects behavior, which affects outcomes. Mediation analysis asks how much of an effect operates through a specified pathway. But natural direct and indirect effects require assumptions about cross-world counterfactuals and mediator-outcome confounding that may be difficult to defend. Interventional analogues or randomized encouragement designs may provide more credible alternatives.

The architecture therefore does not require a single numerical decomposition. It requires the researcher to state the pathway and distinguish a total effect from a mechanism-specific effect. Process tracing, qualitative evidence, and institutional records may strengthen a mechanism claim even when a mediated effect is not point-identified.

5.5 Transportability

Comparative law routinely transports claims. A reform succeeds in Chile and is recommended for Argentina; an AI-liability rule in the European Union is proposed elsewhere; a judicial innovation from one province is treated as a national model. Transport requires more than external resemblance.

Selection diagrams make differences explicit: legal culture, court capacity, federal structure, professional incentives, data quality, and enforcement may alter causal mechanisms or variable distributions (Pearl and Bareinboim 2014). Some differences can be reweighted; others modify the effect itself. DEPCA treats “jurisdiction” not as a nuisance label but as a bundle of candidate transport modifiers. A result should travel only across the dimensions justified by theory and data.

5.6 Partial identification and honest bounds

Legal data are often sparse, selected, and measured after disputes arise. Point identification may be impossible. Partial identification is preferable to false precision. Manski (1995) shows how weak assumptions can produce informative bounds rather than a single estimate. Sensitivity analysis can quantify how strong unmeasured confounding would need to be to alter a conclusion. Probabilities of causation may likewise be bounded under monotonicity or other assumptions rather than asserted as exact singular probabilities.

An identification-ready research program is not one that always produces a number. It is one that knows when the evidence does not license one.

5.7 Causal discovery and the limits of automation

Algorithms can search for graphs compatible with conditional independencies or exploit assumptions about functional form, noise, temporal order, and invariance (Peters, Janzing, and Schölkopf 2017). They can assist DEPCA, particularly when longitudinal institutional data contain many plausible pathways. They cannot infer legal ontology from data alone.

Markov-equivalent graphs encode the same observed independencies while reversing causally important arrows. Hidden confounding, selection, measurement error, and feedback enlarge the ambiguity. A machine may orient an edge because of an assumed non-Gaussian model or invariance across environments; the resulting direction is conditional on that assumption, not a discovered legal fact.

The appropriate workflow is hybrid. Domain knowledge fixes impossible temporal directions and constitutive relations. Discovery methods generate candidate structures or test residual implications. Institutional records and legal history assess mechanisms. Interventions or natural experiments resolve edges where possible. Competing graphs remain explicit when evidence cannot decide.

This is especially important for language models. A model can propose a fluent DAG by associating concepts in text. Fluency supplies no identification. Any AI-assisted graph must preserve provenance for each variable and arrow, identify disputed edges, and pass human legal and methodological review.

5.8 Prediction as a complementary task

The insistence on causal identification should not devalue prediction. Courts, legislators, firms, and citizens often need calibrated forecasts before mechanisms are fully known. A model may predict injunctions or reform failure from pre-treatment features and be useful for planning. The mistake is to interpret predictive importance as an intervention effect.

DEPCA supports a division of labor. Predictive models estimate risks under the observed policy regime. Causal models estimate changes under defined interventions. A predictor may identify where intervention research is most valuable; a causal design may explain which components should change. Both require temporal validation and protection against leakage, but only the second licenses “would change if set.”

The distinction also clarifies prospective tests in EPT-Law. A preregistered probability followed by an observed event tests calibration and discrimination over repeated predictions. It does not by itself validate the proposed causal mechanism. Mechanism validation requires evidence that rival pathways fail and that manipulation or quasi-exogenous variation operates as predicted.

6. Three Applications

6.1 Constitutional lock-in

Constitutional lock-in is often described as the structural resistance of a legal regime to reform. The construct can be used descriptively, predictively, or causally. DEPCA separates those uses.

For prediction, an index combining precedent density, doctrinal elaboration, professional organization, and enforcement infrastructure may estimate the probability that a reform will survive. Predictive evaluation requires pre-treatment coding, out-of-sample assessment, calibration, discrimination, and attention to data leakage. It does not require that each component be a manipulable cause.

For causal analysis, the question must target an intervention. Suppose the intervention is prospective review by a specialized constitutional chamber before implementation, compared with decentralized review after implementation. The outcome is sustained implementation five years after enactment. Pre-treatment causes may include prior constitutional doctrine, party fragmentation, administrative capacity, economic shock, union density, and reform content. Mediators may include injunction timing, legal uncertainty, employer compliance, and mobilization. The causal estimand is the

average effect of the review architecture among eligible reforms, not the effect of “lowering CLI” as an abstract score.

Feedback then enters. Failed reforms may generate new precedents, professional narratives, and organizational investments, increasing future resistance. Let F_t denote failure at time t , P_{t+1} new precedent density, and R_{t+1} design of the next reform. The pathway $F_t \rightarrow P_{t+1} \rightarrow R_{t+1}$ is a concrete niche-construction mechanism. It predicts that repeated attempts are not exchangeable observations.

A dynamic study can test whether failure changes the state variables that condition later success.

This reframing also disciplines statements about crisis. A negative interaction between crisis and an index may predict lower success in high-score systems. To claim that crisis *causes* additional lock-in requires a treatment definition, temporal ordering, comparison, and control of causes of both crisis exposure and reform outcomes. Crisis may itself be a heterogeneous bundle and may alter proposal selection. DEPCA converts a striking coefficient into a research design rather than a conclusion by adjective.

6.2 Liability for artificial intelligence

AI-related harm presents three distinct causal layers. The technical layer asks whether model, data, interface, deployment, monitoring, or human-use choices affected an output or harm. The organizational layer asks how responsibility and information were distributed among developer, deployer, professional user, platform, and victim. The legal layer asks which causal contributions satisfy the applicable tests of attribution, duty, fault, product defect, burden of proof, and scope of liability.

Consider an erroneous legal filing influenced by a generative system. Candidate causes include training-data limitations, retrieval failure, prompt design, misleading interface confidence, lack of source verification, time pressure, professional review, and adversarial manipulation. Conditioning on the fact that a case reached litigation may select disputes jointly caused by severity and detectability. Studying only reported hallucinations may therefore distort the relation between system design and harm.

A policy-relevant intervention might require source-linked outputs and block filing when primary authority is unavailable. The outcome could be the rate of materially unsupported propositions in filed documents. A cluster-randomized trial across teams, an interrupted rollout, or a stepped-wedge design could estimate the intervention’s effect. Mediation analysis could ask whether improvement operates through increased source inspection or reduced generation of unsupported text. Spillovers are plausible because lawyers learn across teams.

For a particular lawsuit, the question changes. Would the harm have occurred if the source gate had been present? That is a singular counterfactual, not the average effect. Even a high probability of necessity does not establish negligence. The court must still decide who owed the verification duty, whether the precaution was legally required, whether the harm fell within the duty’s scope, and how proof burdens are allocated.

DEPCA also clarifies asymmetric intentionality. A user and a corporate or algorithmic system may operate under different strategic and cognitive architectures. That hypothesis may explain why reputation or disclosure mechanisms fail. It does not identify the effect of a regulation. Intentionality level can be modeled as an effect modifier, mediator, or measurement construct, but its role must be tested rather than inferred from the causal rung.

6.3 Channel opacity and reform persistence

Some legal changes endure because the channels through which affected actors would detect, interpret, and contest them are opaque. The hypothesis is not merely that unread rules persist. It is that limited observability changes mobilization, enforcement, compliance, and selection.

Let C_t be channel opacity, K_t stakeholder knowledge, M_t mobilization, E_t enforcement, and P_{t+1} persistence. A candidate mechanism is:

$$C_t \rightarrow K_t \rightarrow M_t \rightarrow P_{t+1},$$

with enforcement and material salience affecting both detection and persistence. The total effect of opacity differs from the indirect effect through knowledge. Documentary obscurity may also correlate with technical complexity, low economic stakes, or deliberate coalition strategy. Those are alternative causes, not footnotes.

A research design could exploit exogenous changes in publication or notification. If an official gazette introduces searchable alerts for some categories before others, the rollout may estimate the effect of increased observability on challenges, compliance, or repeal, subject to assignment and spillover assumptions. Randomized notification to eligible stakeholders could identify a narrower effect of information. Text complexity itself is harder to manipulate cleanly, but controlled experiments may estimate comprehension rather than institutional persistence.

The application demonstrates why causal architecture adds value to evolutionary explanation. Opacity may increase a reform's memetic fitness by suppressing counter-mobilization, even while reducing behavioral implementation. Persistence and efficacy can move in opposite directions, exactly as Proposition 5 predicts.

6.4 A complete worked specification

Consider a hypothetical federation that changes the publication of administrative regulations from a paper gazette to a searchable digital system with automatic notices. The substantive theory predicts that improved observability reduces channel opacity, increases stakeholder knowledge, accelerates challenges to defective rules, and changes which regulatory variants persist.

The population consists of regulations adopted by national agencies during four years before and four years after rollout. The intervention is eligibility for automatic notice on the legally fixed phased schedule. The comparator is otherwise eligible regulation not yet covered. Time zero is publication. Outcomes are: stakeholder access within thirty days, formal challenge within one year, confirmed compliance within two years, and continued legal force at four years. The primary estimand is the intention-to-treat effect of notice eligibility on each outcome. A secondary mediated estimand concerns the pathway through documented stakeholder access.

The favored graph includes agency type, subject salience, affected-population size, technical complexity, and prior controversy as pre-treatment causes of rollout timing or outcomes. Actual notice opening mediates eligibility. Mobilization follows knowledge. Enforcement capacity affects compliance and persistence. A rival graph proposes that agencies strategically phase salient regulations earlier, so rollout timing is confounded by anticipated controversy.

Identification depends on the rollout rule. If categories were assigned by a preexisting technical threshold, regression discontinuity may be plausible. If timing was administrative, difference-in-differences requires conditional parallel trends and no strategic anticipation. If individual recipients

within eligible categories were randomized to reminders, that experiment identifies the effect of reminders but not the full digital-publication reform.

Interference must be addressed. Organizations receiving notices may alert nonrecipients, and a successful challenge can affect similar regulations. Exposure could therefore be defined at professional-network or regulatory-family level. Alternatively, the estimand can be the effect of saturation within clusters.

The evolutionary claim is a separate second-stage question. If notice changes the retention of variants, the analyst must define variant properties and test whether selection differs by observability. A rise in challenge rates alone shows an operational effect, not a shift in memetic evolution. The legal claim is separate again: greater challenge does not establish that challenged rules are invalid. Validity depends on substantive and procedural law.

The permissible conclusion might be: “Under the phased-rollout assumptions, automatic-notice eligibility increased timely stakeholder access and challenges in the covered regulatory population.” A stronger conclusion, “opacity causes unlawful regulations to survive,” remains blocked unless unlawfulness is adjudicated, persistence is measured, and the causal path is identified. This worked example illustrates the function of a claim ceiling: the model permits a useful conclusion while preventing three adjacent overclaims.

7. From Causal Attribution to Legal Judgment

7.1 The normative bridge

Legal doctrine frequently uses causal language: factual cause, proximate cause, adequate causation, material contribution, loss of chance, intervening cause. These are not interchangeable with a scientific estimand. Some are evidentiary standards; some delimit responsibility; some combine factual and normative judgment. The variety of causal questions in adjudication and their relation to responsibility long predates modern graphical methods (Hart and Honoré 1985). DEPCA does not displace that jurisprudence; it makes the empirical premises entering it more explicit.

DEPCA therefore requires a normative bridge with at least five entries: the causal finding, the governing legal source, the bridge premise, the authorized decision maker, and the permissible conclusion. For example, an estimated reduction in unsupported filings under a verification gate is a causal finding. A professional-conduct rule establishes a duty. A bridge premise connects the precaution to the duty’s scope. A court or disciplinary body applies the standard of proof. The permissible legal conclusion follows only after all layers are supplied.

7.2 Burdens and probabilities of causation

Pearl’s analysis of probabilities of necessity and sufficiency offers tools that may illuminate legal attribution (Pearl 2009). The probability of necessity asks, roughly, whether an outcome that occurred under exposure would not have occurred without it. The probability of sufficiency asks whether exposure would produce the outcome in an otherwise unexposed case. These quantities can map onto legal intuitions, but they are not self-executing legal rules.

Identification may require exogeneity, monotonicity, consistency, or experimental and observational information. Legal systems may accept proof below scientific identification, shift burdens because evidence is controlled by one party, or reject liability for policy reasons despite factual contribution. Conversely, a rule may impose strict liability without requiring fault. Causal

models can expose what is known, assumed, and bounded; they cannot legislate the burden of persuasion.

7.3 Validity and evolutionary success

Evolutionary success has no necessary normative valence. A discriminatory doctrine may be stable, transmissible, and institutionally entrenched. A rights-protective innovation may initially be fragile. Treating persistence as validation would reproduce the naturalistic fallacy and misunderstand legal validity.

EPT-Law is most useful normatively when it predicts implementation conditions and unintended consequences while leaving reasons for adoption to constitutional, moral, and political argument. The theory can explain why a rule survives and still condemn it. It can support a desirable intervention and still predict its failure.

7.4 Explanation, reasons, and institutional agency

Legal decisions are produced by agents who give reasons. A causal account that ignores reasons can be incomplete; one that treats stated reasons as transparent causes can be naive. Judicial opinions, agency memoranda, and legislative debates are both actions within a causal process and strategic records created for audiences.

DEPCA treats reasons as variables whose role must be specified. A legal reason may mediate the effect of text on decision, signal coalition alignment, legitimize an independently chosen outcome, or shape future citation and professional learning. Content analysis can measure reasons; process tracing can test their use; experiments can study their persuasive effects. None of those methods assumes sincerity by definition.

Institutional agency also differs from individual intention. A corporation or court can display stable decision procedures without possessing a unitary mind. Causal analysis may model organizational rules and information flows directly. Intentional-language explanations remain useful at a predictive stance, but responsibility requires identifying the legal persons, officials, or governance structures to which doctrine attributes acts and duties.

This point links DEPCA to asymmetric intentionality without collapsing the frameworks. Different agents may reason at different recursive levels, and institutions may aggregate them through procedures that produce another pattern. The causal question concerns how those architectures affect outcomes under interventions. The legal question concerns when their outputs are attributable to an actor.

8. An Empirical Protocol for DEPCA

8.1 Step 1: State the question before collecting variables

Every study should begin with a one-sentence causal contrast. It must identify population, intervention, comparator, outcome, and horizon. “The effect of constitutional lock-in” fails this test. “The five-year effect of prospective centralized review, compared with decentralized post-enactment review, on sustained implementation of national labor reforms enacted in presidential democracies from 1990 to 2030” passes as a target, even if later judged infeasible.

The paper should then classify whether the goal is association, prediction, intervention, mediation, transport, or singular counterfactual attribution. A study may answer more than one, but each requires separate claims.

8.2 Step 2: Freeze time zero and eligibility

Time zero is when units become eligible, treatment is assigned or begins, and follow-up starts. Misalignment creates immortal-time and selection biases. In reform studies, proposals that never reach enactment cannot silently enter the same risk set as enacted laws unless the estimand concerns legislative adoption. In litigation studies, cases selected after an outcome cannot represent all disputes.

Eligibility rules should be prespecified. Reforms, jurisdictions, cases, or agencies should not be included because their outcomes are already known and theoretically convenient.

8.3 Step 3: Draw at least two plausible graphs

The primary graph should represent the favored mechanism. A rival graph should encode the strongest alternative, such as coalition power causing both institutional structure and reform success. Where graphs imply different adjustment sets or observable independencies, the study can discriminate. Where they are observationally equivalent, the uncertainty must remain.

Every arrow should have a source type: established literature, legal-institutional knowledge, direct evidence, or hypothesis. The graph is a ledger of assumptions.

8.4 Step 4: Define measurement models

Indices must be constructed from variables measured before the relevant outcome. Coding manuals, inter-rater reliability, missingness, and measurement invariance across jurisdictions should be reported. If a construct changes meaning across legal systems, pooled comparisons may be invalid even when numeric scales match.

The canonical CLI formula within the current EPT research program should remain fixed when that construct is used: $CLI = 0.30P + 0.25D + 0.20O + 0.25E$. Alternative component sets or weights must be named as different models rather than silent versions. More importantly, empirical studies should test whether components have separable causal roles instead of assuming the weighted sum is an intervention.

8.5 Step 5: Declare the identification strategy

The protocol must state why the chosen design identifies the target estimand. For adjustment, list the set and justify it from the graph. For difference-in-differences, report pre-trends, anticipation, treatment timing, heterogeneity, and spillovers. For instruments, defend relevance, independence, exclusion, and monotonicity. For longitudinal g-methods, define histories and treatment models. For mediation, state cross-world or interventional assumptions.

If no route is defensible, the study remains descriptive, predictive, or theoretical. That is a valid output.

8.6 Step 6: Precommit falsification and sensitivity tests

At minimum, identify a negative-control outcome or exposure where plausible, a placebo timing test, an alternative outcome definition, and a sensitivity analysis for unmeasured confounding. Institutional studies should test alternative lags. Comparative studies should report leave-one-jurisdiction-out results. Index studies should test component ablation and temporal leakage.

Falsifiability should not be reduced to assigning a probability and waiting. The predicted observation must distinguish the model from a rival and define what would count against the mechanism.

8.7 Step 7: Separate estimation from transport

After estimating an effect in a source population, list the variables whose distributions or mechanisms differ in the target. State which can be standardized and which threaten structural transport. Claims should be narrowed to populations supported by the model.

8.8 Step 8: Apply the causal-normative bridge

Only after the causal result is stated should the paper ask what legal conclusion follows. Cite the positive-law source, identify the bridge premise, and disclose residual discretion. For policy recommendations, distinguish predicted consequences from the normative criterion used to value them.

8.9 Minimum reporting table

Every causal EPT-Law paper should report:

Element	Required statement
Causal rung	Association, intervention, mediation, transport, or counterfactual
Population and time	Eligible units and temporal window
Intervention	Executable contrast and versions
Outcome	Measure and horizon
Estimand	Formal target quantity
Graph	Variables, arrows, and rival structure
Identification	Design and assumptions
Measurement	Construct-to-indicator mapping
Robustness	Falsification, sensitivity, and bounds
Transport	Supported target populations
Normative bridge	Legal source and additional premise
Claim ceiling	Strongest wording justified

9. Rival Accounts, Risks, and Limits

9.1 Is EPT necessary?

A critic may accept DEPCA while rejecting evolutionary terminology. Institutional economics, historical institutionalism, organizational sociology, and ordinary causal inference can model feedback, path dependence, and mediation without replicators or phenotypes. This objection is serious.

EPT earns its place only if it contributes discriminating hypotheses. These may concern the mechanisms of high-fidelity transmission, the divergence between replicator fitness and social welfare, institutional structures that improve a norm's own reproduction, or selection across competing variants. If the same model and predictions arise without the evolutionary vocabulary, EPT is interpretive compression rather than an indispensable causal theory.

DEPCA makes that vulnerability testable. Researchers can compare an EPT graph with coalition, efficiency, cultural-attraction, or state-capacity graphs. The evolutionary account should survive because of explanatory and predictive gain, not because all institutional continuity is redescribed as replication.

9.2 Cultural attraction and reconstruction

Cultural-attraction theory disputes a picture of culture dominated by faithful copying. Legal texts are copied with unusual fidelity, but their meanings are reconstructed by courts, agencies, and professions. A doctrine may converge toward an attractor because of cognitive, moral, or institutional dispositions rather than because a stable meme reproduces intact.

DEPCA can represent both accounts. A replicator model predicts path dependence around identifiable lineages and transmission channels. An attraction model predicts convergent reconstruction even without common ancestry. Citation genealogy, textual similarity, independent emergence, and experimental interpretation can help distinguish them. The architecture therefore narrows rather than suppresses the rival.

9.3 Manipulability and constitutive concepts

Some legal properties are constitutive rather than readily manipulable. Sovereignty, validity, or constitutional identity may not admit a simple intervention. Forcing them into $do(X=x)$ can distort the question. The remedy is not to abandon causal reasoning but to intervene on institutional practices, rules, information, or procedures that instantiate the broader concept. When no coherent intervention exists, the claim may remain mechanistic, historical, or constitutive rather than causal in the interventionist sense.

9.4 Interference and system-level effects

Legal interventions affect actors beyond treated units. A precedent changes expectations across cases; one jurisdiction's rule changes investment or migration elsewhere; professional learning crosses organizational boundaries. The stable-unit-treatment-value assumption often fails. Studies should define exposure mappings, clusters, networks, or system-level estimands. Individual-level estimates cannot simply be aggregated when spillovers are part of the legal mechanism.

9.5 Data generation by institutions

Legal data are not passive traces. Institutions decide what is recorded, published, appealed, cited, and digitized. Citation networks reflect publication and strategic citation. Reported AI incidents reflect detection and incentives. Observed enforcement reflects capacity and selection. The data-generating process belongs inside the causal model.

This problem is acute for historical corpora and cross-jurisdictional research. Missing records are rarely random. Channel opacity can determine both the phenomenon and whether it is observed. DEPCA therefore requires a measurement and selection graph, not only a substantive graph.

9.6 Formalization without false authority

Equations can create unwarranted confidence. The structural functions in this paper formalize dependencies and questions; they are not estimated laws. The six propositions are analytical constraints, not empirical findings. The applications are worked specifications, not tests. Any later manuscript must resist importing the language of proof from formal coherence into claims about the world.

10. Research Agenda

The first priority is retrospective reclassification of the existing EPT-Law corpus. Each paper's major claims should be coded as descriptive, predictive, intervention, mediation, transport, or counterfactual. Numerical findings should be linked to data and estimands; theoretical constructs should be labeled accordingly. This audit would expose where prior language exceeds identification and where existing evidence can support a more precise claim.

The second priority is prospective validation of constitutional lock-in components. Instead of seeking only a larger cross-country correlation, future studies should identify interventions that alter precedent, review procedure, professional organization, or enforcement capacity. Natural experiments and staggered procedural changes may offer leverage. Out-of-sample prediction remains valuable but should not be called causal validation.

The third priority is longitudinal analysis of repeated reform. Dynamic treatment regimes can test whether failed attempts alter the institutional state and reduce or increase later success. This is the causal core of niche construction: interventions change the environment in which subsequent interventions operate.

The fourth priority is mechanism research on legal AI. Source gates, interface warnings, audit logs, adversarial controls, and allocation of review duties are manipulable components. Experiments and phased rollouts can estimate their effects. Legal responsibility should then be analyzed through a separate normative bridge.

The fifth priority is transport. EPT-Law's comparative claims should identify which mechanisms are expected to remain invariant and which depend on jurisdiction. A theory of legal evolution becomes stronger when it predicts its own scope conditions.

10.1 Distinctive and falsifiable predictions

DEPCA is worthwhile only if it generates tests that differ from generic claims that history matters. Five predictions define an initial boundary.

Feedback accumulation. When a failed reform creates precedents, organizations, or professional narratives that are later used against similar reforms, the conditional probability of subsequent failure should rise even after accounting for stable preexisting institutional features. A simple static-rigidity account predicts persistence but not necessarily this attempt-specific state change. Evidence that failures leave no measurable institutional trace would count against the niche-construction pathway.

Pathway substitution. If a legal replicator operates through an institutional phenotype, disabling one transmission pathway should shift activity toward substitute pathways when the memplex is robust. Removing an administrative rule may increase reliance on litigation, private standards, or professional discipline. A pure text-effect model predicts attenuation without systematic substitution. The prediction can be tested with interrupted network and event-history data.

Fitness-efficacy divergence. Some variants should become more persistent while producing less implementation, and others should have strong behavioral effects while becoming politically fragile. Finding a stable monotonic relationship between retention and desired outcomes across environments would weaken Proposition 5 and reduce the added value of distinguishing fitness from efficacy.

Observability reversal. Increasing channel observability should not have a uniformly positive effect on persistence. It should increase implementation when ignorance is the main barrier and decrease persistence when opacity suppresses opposition. The sign should therefore depend on whether information primarily activates beneficiaries, enforcers, or challengers. A context-insensitive observability effect would contradict the proposed selection mechanism.

Orthogonality under intervention. Interventions that increase an agent's deliberative or intentional sophistication need not improve the identification of the intervention's effect, while methodologically stronger designs need not alter agent cognition. Experiments that manipulate explanation or reflection can test behavioral consequences; random assignment determines identification. If those dimensions systematically coincide only because the same intervention changes both, the relation is empirical and local, not conceptual identity.

Each prediction includes a possible adverse result. This matters. A research programme progresses when auxiliary architecture exposes the core to discriminating tests, not when every observation can be assimilated after the fact.

11. Conclusion

The Extended Phenotype Theory of Law begins with a causal intuition: legal replicators can reach beyond texts and minds into institutions that alter behavior and reproduce the norms that helped build them. Niche construction adds a second intuition: those institutions alter the selective environment facing later variants. Together they describe a dynamic system of considerable explanatory promise.

Promise is not identification. A vocabulary of cause, mechanism, feedback, and selection can remain compatible with many rival histories. The next stage of the research program must specify what is observed, what is intervened on, what is imagined otherwise, and under which assumptions each question can be answered.

DEPCA supplies that discipline. Temporal unrolling preserves reciprocal causation while requiring causal order. Institutional role differentiation prevents indiscriminate adjustment. Measurement non-identity keeps indices from becoming reified causes. Orthogonality prevents causal and intentional ladders from collapsing into one another. Persistence-efficacy separation protects evolutionary claims from circularity. Causal-normative separation preserves the authority of law without insulating factual claims from scientific scrutiny.

The consequence is not a narrower theory. It is a theory capable of being wrong in informative ways. A causal architecture tells researchers what data to collect, what comparisons to seek, what assumptions to defend, what rival explanations to test, and where a conclusion must stop. That is how an evolutionary account of law moves from metaphor to mechanism, and from mechanism to identification.

Lakatosian Epistemological Declaration

This article treats EPT-Law as a developing research programme in the Lakatosian sense (Lakatos 1978). The claim that legal norms may operate as cultural replicators and generate institutional extended phenotypes belongs to the programme's theoretical core. DEPCA is part of its revisable methodological belt. Its value depends on whether it produces novel, risky, and independently testable implications rather than post hoc accommodation. The six propositions in this article are conceptual constraints. Their empirical applications require separate designs and evidence. No simulation or causal-effect estimation was executed for this paper.

AI Assistance Disclosure

AI tools were used to assist with source organization, structural drafting, consistency review, and document production. The author selected the thesis, determined the legal and theoretical arguments, reviewed the sources and claims, and assumes full responsibility for the manuscript. No AI-generated statement is offered as an independent source of evidence.

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